

Tucker Vana

Engineering case packet

My work combines technical planning with hands-on setup, operation, and repair. These cases show specific decisions and my contribution.

CASE 01 / Pouch Factory

Production air supply

Air supply selected for initial production, with allowance for a planned second line and a constrained startup budget.

First line supplied

CASE 02 / Fabrication / CNC

Waterjet pump rebuild

An intermittent cutting fault led to pump inspection, a standard scheduled rebuild, and resumed production.

3 working days

CASE 03 / Maybell Quantum

Prototype instrumentation

Instrument readout integration and gas-valve controls shared a GUI for the team's refrigeration prototype.

One operator interface

B.S. Mechanical Engineering, Colorado School of Mines, December 2021.
Cum laude; 3.65 GPA.

Earlier practical work: truck driving, forklift/skid-steer operation, material handling, and yard work at DBC Irrigation Supply.